Lab Minheap

Data Structures CSCI 2320

Lab Objective

Learn how to implement and use a Heap ADT in C++.

Lab Overview

You will receive a partial implementation of a Heap in C++. Your mission is to implement some of the methods for the class as well as understand all the other methods.

Lab Tasks

Implement the following methods for the Heap class in `MinHeap.h`:

- `void insert(int val);` This method add a value to the heap per heap rules.
- `int peek();` This method returns the value at the top of the heap.
- `int extractMin();` This method removes and returns the value at the top of the heap.

Final output

Your program must compile and produce output.

Rubric

Name	Description	Points
Coding Style	Run cpplint on student code	10
MinHeap Starts Empty	Heap initializes with size 0	5
MinHeap Insert Single Value	Inserting one value updates top and size	5
MinHeap Peek Smallest	`peek()` returns the smallest value after inserts	10
MinHeap Extract Sorted	`extractMin()` returns values in ascending order	15
MinHeap Duplicate Values	Correctly handles duplicate values	10
MinHeap Negative Positive Values	Correctly handles negative and positive values	10
MinHeap Peek Empty Throws	`peek()` on empty heap throws `std::out_of_range`	10
MinHeap Extract Empty Throws	`extractMin()` on empty heap throws `std::out_of_range`	10
MinHeap Interleaved Size	Size tracks correctly across interleaved operations	10
MinHeap Stress Order	Maintains min-heap order under stress inserts/extracts	5
Total Points		100

Due Dates and Honor

The due date is specified on Blackboard.

This is an ***independent*** programming project, and it is very important that you understand and abide by the ***academic integrity policy*** concerning programming projects. Remember, your personal honor and integrity is far more important than your grade on the project.

Grading

This lab is available in GitHub Classroom. Accept the URL on Blackboard and then clone your repository to your machine for development. Your lab will be graded automatically via GitHub. Please check the grading results each time you check in your code. Your final grade will be based upon your last sync to GitHub before the deadline.

Project Artifacts

The following should be completed by the due date/time specified on Blackboard.

- Check in all source code changes to your GitHub repository. Please check your URL using a web browser to verify that your changes have been synced.
- Submit the URL for your repository to Blackboard.

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